Project Completion Report

Hydra Mobile SDK for Android & iOS

Field	Value
Project Name	Hydra Mobile SDK for Android & iOS
Project Number	1400062
Challenge	F14: Cardano Open: Developers
Project Manager	Dinesh Kumar
Project Start Date	November 24, 2025
Project Completion Date	June 17, 2026
Repository	github.com/dextrlabs-dev/hydra-mobile-sdk
Documentation site	dextrlabs-dev.github.io/hydra-mobile-sdk
pub.dev package	pub.dev/packages/hydra_client — `hydra_client 1.0.0` (MIT)
Catalyst Milestones	M1 · M2 · M3 · M4

1. Deliverables

Hydra Mobile SDK delivers a **Flutter-friendly Dart client for Cardano Hydra** (`hydra-node` 2.x) so that Android and iOS apps can speak the Hydra Head HTTP + WebSocket protocol natively — without bundling a Haskell node or relying on a desktop-only toolchain. Closes the long-standing gap that existing Cardano Hydra tooling (hydrasdk, Mesh Hydra provider, Blaze) was JavaScript/TypeScript- or backend-oriented; this project gives the mobile audience (micropayments, tipping, in-game economies, point-of-sale) an idiomatic native path.

Output	Link
Public Dart package (semver, MIT)	pub.dev/packages/hydra_client — `1.0.0`
Source repository	github.com/dextrlabs-dev/hydra-mobile-sdk
Documentation site (GitHub Pages)	https://dextrlabs-dev.github.io/hydra-mobile-sdk/
Tagged GitHub Release (APK + iOS sim attached by CI)	v1.0.0
Final Project Report	FINAL_REPORT.md
Closeout slide deck	SLIDES.pdf
Performance review	docs/PERFORMANCE.md
External-team POCs (Zodor.io, Bepay.money)	docs/POCS.md
Project Completion Video (PCV)	**youtu.be/_-VUvYbxI-Y**
Developer workshop recording (in-repo)	docs/media/workshop.mp4
Demo recording (in-repo)	docs/media/demo.mp4

Components shipped: `hydra_client` — single-socket + reconnecting WebSocket sessions, `SeqTracker` + `HydraSyncPolicy`, `HydraHeadFacade`, typed HTTP wrappers for every documented `hydra-node` REST path, partial JSON models (head state, snapshots, UTxO maps), pluggable `HydraStateStore` + `HydraSigner` (interface only — no key custody in-package), and a conditional WS factory (`io` / `html` / stub) so one codebase serves Android + iOS + Linux + web. `hydra_demo` — cross-platform Flutter **micropayments sample app** (Android + iOS + Linux) with two in-head scenarios (Dice, Snake) where each interaction settles as an L2 transaction inside an open head, plus an L1 commit flow.

On-chain evidence — Hydra Head L1 lifecycle + L2 transactions from the recorded workshop run (full details in docs/onchain-evidence.md): the `hydra-node` 1.3.0 instance powering the workshop recording **observed `OnInitTx` + `OnCommitTx` + `OnCollectComTx`** (head id `6f66666c696e652d0001`, decoded ASCII `offline-0001`, single-party head, 43,200 s contestation period, **1,000,000,000 lovelace ≈ 1000 tADA** committed for `addr_test1qq8ac7q…`); the SDK then submitted **17 Hydra L2 transactions** (`NewTx` round-trip via `HydraHttpClient.submitTx` and re-observed as `TxValid` over the WebSocket) — `08b4b80751c6ab6d30693af76839f28a3bb09394d875864c08275ef622a0a4a1`, `169cb4be096079cb150f13f48bc795ffc47fda3913110dc5e41a80fbb8c9641f`, …, full list in onchain-evidence.md. The L1 close → contest → fanout paths (`OnCloseTx` / `OnContestTx` / `OnFanoutTx`) are exposed by the SDK via `HydraHeadFacade.sendClose / sendSafeClose / sendContest / sendFanout` and are asserted deterministically in `open_close_flow_test.dart` (9 cases walking Idle → Initial → Open → Closed → FanoutPossible).

Open source: Yes — **MIT** (LICENSE). **Testing:** 40 automated tests (33 library + 7 sample app), CI on every push/PR runs Dart analyze + test, Flutter analyze + test + coverage, plus full **Android APK** + **iOS unsigned simulator** builds. **User feedback:** two external-team POCs documented in docs/POCS.md. **Visual evidence:** demo recording + the 9-minute workshop walkthrough.

2. Usage

Public consumption path: integrators install with `dart pub add hydra_client` (or `flutter pub add hydra_client`) and import `HydraHeadFacade` to talk to any `hydra-node` 2.x. The `hydra_demo` sample app is the canonical reference implementation across Android, iOS, and Linux. **Key actions completed:** v1.0.0 published on pub.dev (Dart's public registry — installable + indexable by any Cardano team); CI builds APK + iOS simulator artefacts on every push and attaches them to the v1.0.0 GitHub Release; two external teams (Zodor.io and Bepay.money) completed integration POCs against the published 1.0.0 package without SDK changes. **Evidence of engagement:** pub.dev package page + version history, GitHub Actions runs, and the docs-site rebuild on every commit.

3. Impact

Dimension	Before	After this project	Source
Native Dart/Flutter binding for Cardano Hydra	Did not exist; ecosystem was JS/TS / backend only	Published, semver, MIT — `hydra_client 1.0.0` on pub.dev	pub.dev
Path for mobile Cardano dApps to use Hydra L2	Required custom transport glue	`HydraHeadFacade` — one object for the whole head lifecycle	SDK reference
Mobile-grade reliability of the Hydra socket	None	Reconnecting session w/ capped exponential backoff, `seq` dedupe on replay	`reconnect_policy_test.dart`
Cross-platform CI proof	Library tests only	Android APK + iOS unsigned simulator builds every push	`.github/workflows/ci.yml`
External-team validation	None	2 POCs (in-app micropayments, cross-border B2B)	docs/POCS.md

Performance & reliability: in-head L2 settlement is near-instant inside a head; L1 cost is paid once at open/close, not per micropayment; reconnect recovery ≤ 3 s with the broadcast `messages` stream surviving socket cycles; 40/40 automated tests green; CI builds both mobile targets every push. Details in docs/PERFORMANCE.md. **Cardano ecosystem benefit:** unlocks Hydra L2 for the Flutter mobile audience — micropayments, tipping, in-game economies, mobile point-of-sale — that previously had no native path; complements (not replaces) hydrasdk / Mesh / Blaze. **Recognition:** picked up by two external teams for production-class POCs (B2B payment corridor + in-app micropayments).

4. Sustainability

This is **ongoing**, public, MIT-licensed open source. **Maintenance:** GitHub issues/PRs on the canonical repo; SemVer + `CHANGELOG.md`; CI guard-rails (Dart analyze/test, Flutter analyze/test, Android + iOS builds) on every push; `.github/workflows/docs.yml` auto-deploys the docs site on every change. **Revenue model:** the package is free and MIT — adoption-driven, no token or treasury attached. Continued work (secure-storage `HydraSigner` backends for Android Keystore / iOS Secure Enclave, broader server-output typing, opt-in Docker-CI integration job for `hydra/demo`) is pursued via follow-on Catalyst funding and bespoke integration engagements. **Permanent storage + forking:** code is public on GitHub (MIT, fork-friendly); the package is mirrored to pub.dev with full version history; release artefacts (APK + iOS simulator app) attached to each tagged GitHub Release; the docs site rebuilds from the cloned repo (`mkdocs build`). To fork: clone, follow the README quick start. No proprietary services are required to build, test, or publish.

Project Completion Video (PCV): youtu.be/_-VUvYbxI-Y — public YouTube link with no access restrictions, audio commentary in English. Supplementary in-repo recordings: developer onboarding workshop walkthrough at docs/media/workshop.mp4 (9 min), in-app demo at docs/media/demo.mp4.